

Explain observable properties of solids, liquids and gases by modelling the motion and arrangement of particles

Learning intention

- explain observable properties of solids, liquids and gases by modelling the motion and arrangement of particles.

Teach and model

- exploring examples that demonstrate that gases have mass.
- exploring, through guided discussion, ideas about what is between particles.
- classifying substances as solids, liquids and gases and investigating their properties.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.